HealthConnect Rwanda Analytics Report

HealthConnect can use these records to review appointment volume, collected payments and feedback. Comparisons across facilities need care: most collections and feedback responses do not link to the appointment extract. The extract also does not record attendance outcomes, so it cannot establish a no-show rate.

Source Figures for the Executive Dashboard

These are the expected values with filters cleared. Each figure comes from its own source and describes a different population.

Measure	Source value	What it measures
Registered patients	**20,000**	Distinct IDs in the patient register.
Recorded appointments	**5,000**	Distinct appointment IDs in the supplied extract.
Facilities	**20**	Distinct facility IDs in the department master.
Doctors	**350**	Distinct doctors in the staff register. The saved card uses the appointment extract, which contains the same 350.
Reported revenue	**RWF 1,132,677,500**	Sum of 12 monthly Revenue Summary rows.
Patient satisfaction	**2.99 / 5**	Mean rating across 21,064 responses.

What the Week 7 Tests Established

A controlled test showed that joining appointments to a lookup with one row per bed inflated the appointment count by department from 5,000 to 50,000. Joining to one row per department restored the source count. The 50,000 figures came from the test and was not observed on the dashboard.

The Week 7 review recorded 14 passing targeted retests in a Power BI and SQLite compatibility run. The Week 8 review recalculated the source figures and checked the saved Power BI fields. The four on-screen checks in the dashboard validation record still need to be run in Power BI Desktop.

Where Appointment Links Are Missing

The extract records 5,000 appointments across 4,443 distinct patients, dated 1 January to 28 December 2026. The Revenue Summary separately reports 60,809 appointments. Its aggregate does not identify individual appointments or explain what happened to them.

The chart shows how often billing, feedback and laboratory records can be connected to an appointment in the extract. The counts describe matches by AppointmentID.

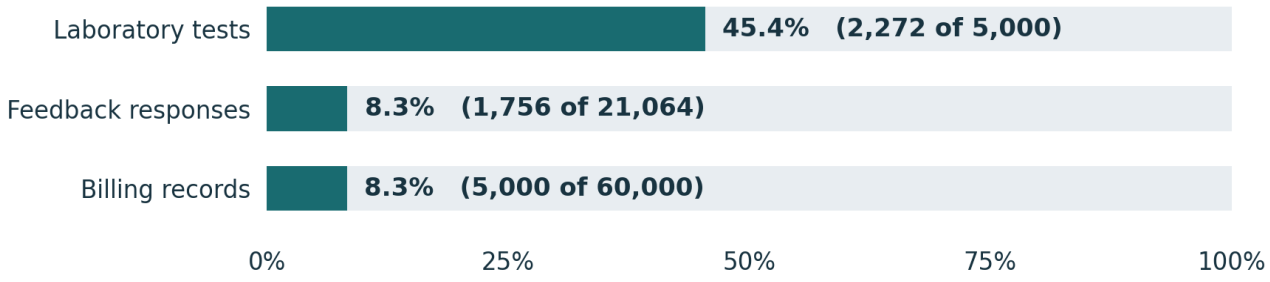


Figure 1. Share of each source that matches a recorded appointment.

Facility feedback comparisons cover only 1,756 responses. For laboratory workload, keep all 5,000 tests in the total and show the 2,728 unmatched tests separately when grouping by facility or department.

Payments and the Unpaid Balance

Measure	Amount
Billed	**RWF 9,309,401,961**
Collected payments	**RWF 8,389,454,556**
Outstanding billing balance	**RWF 919,947,405**
Collections linked to an appointment	**RWF 694,551,320**
Collections without an appointment link	**RWF 7,694,903,236**

Only 8.28% of collections link to an appointment and can be grouped by facility or department. Label those charts as appointment-linked payments. The balance is billed less collected. Missing transaction dates and expenses prevent conclusions about overdue amounts or losses.

How Data and AI Can Improve Patient Support

Project question: HOW CAN HEALTHCONNECT CLINIC USE DATA AND AI TO REDUCE MISSED APPOINTMENTS AND IMPROVE THE PATIENT SUPPORT EXPERIENCE?

Start by recording what happens to each appointment: attended, missed, cancelled or rescheduled. Agree which appointments are eligible for the no-show calculation and establish a baseline. The clinic can then test reminders and easier rescheduling against that baseline, tracking attendance and patient feedback to see whether the changes help.

An AI assistant could answer routine scheduling questions using approved clinic information, explain how to reschedule and refer requests it cannot handle to staff. Once reliable outcomes exist, Data Science could test whether appointment history helps identify bookings at higher risk of a no-show. This analysis did not test an AI assistant or measure the effect of reminders.

Recommended Actions

Improve the appointment export. Capture appointment outcomes and timestamps, and obtain a more complete set of appointment IDs. Reconcile facility and department assignments with the authoritative lookup before comparing sites.

Make payment coverage visible. Keep payments without an appointment link in the overall collection total. Show their share alongside facility charts and keep reported revenue, payments and claims under separate labels.

Measure patient experience over time. Capture feedback submission dates and validate the appointment references so trends and facility comparisons can be based on the right records.

Evidence for the Demonstration

The planned Power BI test is to select HF011, compare the expected 273 recorded appointments and RWF 39,810,993 in linked payments, then clear the selection and check that the baseline returns. These are source benchmarks. The live results and any screenshots should be recorded in the separate validation record.

Data Science could use the attendance field assessment, and Project Management could coordinate a complete export and resolve conflicting records. These are proposed uses. No cross-track exchange or group walkthrough is documented.

Sources: AnalystLab Africa Week 8 project guide, pp. 2 and 6–7; Joyce's Week 7 Testing and Refinement Report; Week 8 KPI and dashboard validation records; supplied appointment, patient, department, doctor, revenue, billing, payments, laboratory and feedback extracts. Currency follows the Week 7 RWF reporting assumption.